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M 303 BOILER FEED PUMPS

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with Part 4, Subsection 1.2 “Summary of Work”.

Three (3) x 50% duty capacity feedwater pumps in each pressure feedwater each HRSG shall be supply for combine cycle power plant.

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The scope of work described hereinafter is for one complete set of feedwater pump in each pressure feedwater system unless stated otherwise.

- M.3.1. Pump
- M.3.2. Electric motor
- M.3.3. Couplings and guards
- M.3.4. Bedplate
- M.3.5. Sealing system
- M.3.6. Minimum flow system
- M.3.7. Insulation and lagging
- M.3.8. Oil system
- M.3.9. Instruments and control equipment
- M.3.10. Electrical equipment
- M.3.11. Foundation for all equipments specified this specification.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of Part 4, Subsection 2.5. Codes and Standards and equivalent.

The codes and standards of the following organizations extracted from Part 4, General Technical Requirements are specifically applicable to the design, manufacture and testing of the equipment included in this Specification :

ANSI	American National Standards Institute
ASME	American society of Mechanical Engineers
ASTM	American Society for Testing and Materials
HIS	Hydraulic Institute Standards
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers

3. **DESIGN AND OPERATING CONDITIONS**

The pumps shall be designed for and shall operate satisfactorily under the following conditions

The feed water system design flow shall be based on the peak combined cycle plant operation plus 10% flow margin.

The pump NPSH requirements, suction piping system and deaerator location shall be considered as a whole in order to have pump operation free of cavitation even during extreme transient conditions (i.e. load reduction or full load trip of turbine). Special attention shall be given to first stage impeller design. NPSH data of pump shall be based on 1 % head reduction across the first stage impeller.

The pumps shall be designed to handle water with a pH of 7.0 to 10.0.

The boiler feed pumps shall be arranged to draw feedwater from the deaerator and discharge into the HRSG economizers.

Automatic facilities shall be provided to enable each pump to act in duty or standby modes and to enable the standby feed pumps to come into service on falling pressure in the respective discharge main or failure of an operating pump.

The run up time for the standby pump following a trip of the running pump shall be stated in the proposal.

The Bidder shall submit the feedwater system hydraulic characteristic curves upon which the feed pump characteristics are superimposed for each pressure feed systems. The drawings shall indicate clearly the anticipated normal operating and emergency conditions and also include:

- Each pressure feedwater system with one or more pumps in operation and deaerator pressure decay
- Each pressure steam bypass in operation
- Each pressure steam system safety valves at full lift

3.1 **Other Design Requirements**

The pumps shall be designed in accordance with the General Technical Requirements. It is anticipated that the boiler/turbine units shall over their useful life operate as base load and variable load plant in sliding pressure mode. Therefore the pumps installed must be capable of meeting on a continuous basis all these conditions of operation.

The pumps provided shall have matched characteristics and shall be suitable for parallel operation.

The design total head requirement of the pumps shall be calculated under the following conditions

- Deaerator minimum operating level and rated HRSG pressure
- Normal water level in boiler drum
- The maximum normal operating pressure at the superheater outlet and associated system friction including superheater pressure drop at 105% HRSG rating, or the highest accumulated safety valve setting pressure and associated system friction at 100% HRSG rating whichever is the greater.
- No allowance for drop in frequency of electrical power supply need be made provided the pumps are able to generate the head required at boiler rated pressure when the frequency changes to 50 ± 2.5 Hertz.
- The feed water control valve characteristic and pressure drops shall also be taken into consideration.

- The flow shall be calculated considering the plant operating conditions.

The Bidder shall consider effect of deaerator pressure decay due to steam supply failure on boiler feed pump suction conditions.

The pumps shall be of suitable design to withstand any possible thermal shock (either hot or cold) without damage or leakage. The Bidder shall describe the design -features which enable the pumps to withstand thermal shock.

The manufacturer shall meet all applicable requirements of Section VIII, Division I of the ASME Boiler and Pressure Vessel Code or approved standard, for calculating thickness of pressure retaining parts and in sizing cover bolting.

4. DETAILED REQUIREMENTS

4.1 Pump

Each pump shall be of the ring section, horizontal, centrifugal, multistage, hydraulically balanced type.

Main parts of the pump shall be ruggedly designed and constructed to withstand continuous heavy duty service over the life of the power plant.

The design pressure for each pump casing shall be at least the maximum pressure which can be generated with cold water at maximum rated speed of the pump and with the maximum possible suction pressure.

Pump suction, discharge and intermediate take off nozzles shall be integral with or connected to the casing.

Lifting eyes or lugs shall be provided as required.

Casing shall be arranged for easy removal of impellers, diffusers etc.

All special tools and equipment necessary for the correct tightening of the casing bolts shall be provided.

Means shall be provided to ensure correct alignment during expansion and when reassembling.

The pump casing shall be capable of transmitting all forces and moments, imposed by the pipework, to the equipment foundation without distortion or affecting pump operation.

The Bidder shall give written confirmation to the Owner that the forces and moments imposed on the pump casing by the associated pipework and fittings shall be within the limits acceptable to the pump manufacturer.

Bidder shall furnish pressure break down orifices and balance chamber leak-off flow monitoring orifices. Warm orifices if required shall be provided.²

Vent and drain valves shall be furnished as required.

4.2 Shaft and Impellers

The pump shaft shall be of forged steel with corrosion resistant material. Impellers shall be fitted on the shaft in such a manner to be readily removable and to ensure freedom from thermal distortion

Each impeller shall be balanced individually and the entire rotating element shall be statically and dynamically balanced.

Any unbalanced axial thrust within the pump shall be carried by an approved type of balancing device, a balance drum.

4.3 Wearing Rings

Hardened wear rings and bushings designed for severe service shall be provided on casings and where otherwise required. The rings and bushings shall be easily replaceable, shall reduce leakage to acceptable levels and shall be designed such that no leakage will occur between the ring or bushing and part to which it is affixed. All wear rings and bushings shall be securely locked in place by means of lock screws or other proven device.

4.4 Balancing Device

An internal hydraulic balancing device drum type shall be provided to counteract the impeller's axial thrust.

4.5 Shaft Gland Seals

The Bidder shall offer mechanical seals as the main offer.

The Bidder shall provide details of the mechanical seals and associated supporting systems in the Proposal.

Mechanical seal cooling water shall be cooled by means of a tubular heat exchanger on the closed cooling water system.

Special consideration shall be given to the mounting and supporting of the cooler tubes to allow for the effects of expansion and vibration. Water from the cooling water system shall pass through the tubes of the heat exchanger, and the mechanical seal cooling water through the shell and over the tubes.

Full details of mechanical seal maintenance and inspection procedures shall be stated in the Proposal including

- the expected life of the seal faces
- inspection facilities for monitoring seal condition, and frequency of inspection
- detailed seal replacement procedures including man-hour requirement and overall time required

4.6 Bearings and Bearing Pedestals

Radial bearings shall be of the sleeve type, forced feed oil lubricated bearings.

Shaft bearings shall be of the sleeve type forced feed oil lubricated bearings arranged to be readily accessible for inspection and maintenance purposes.

A forced feed oil lubricated thrust bearing shall be provided sized to carry the maximum unbalanced shaft thrust during start up and run down.

Bearing pedestals shall be cast or fabricated from steel. Cast iron or nodular iron bearing pedestals are not permitted.

4.7 Lubricating System

A separate lubricating system shall be provided for each pump. The system shall consist of a main oil pump directly driven from the pump shaft, electrically driven auxiliary lube oil pump, oil sump, water cooled oil cooler, strainers, filters, pipework and all pressure gauges, sight flow indicators, pressure switches and valves, etc., necessary for the safe start up and operation of the unit.

An oil cooler shall be provided for each pump sized so that all oil temperatures are maintained below the maximum working oil temperature as recommended by the oil manufacturer. The cooler shall be arranged in the oil system so that the oil pressure in the cooler is always greater than the cooling water pressure.

The Bidder shall state in the Bid the proposed procedure for ensuring that an adequate supply of lubricating oil list to all bearings is available prior to starting a feed pump.

Oil piping shall be furnished complete with pressure and temperature gauges, sight flow indicators and flexible connectors as required.

Piping shall be arranged to permit components on the pump to be removed without disassembling piping.

Piping shall be designed according to Part 4, Technical Specification M-001 Piping.

Oil coolers shall have ample capacity to cool to the desired temperature all of the oil required when supplied with the proper amount of cooling water at the conditions specified. in Subsection 1.2. Design of the oil cooler shall be such that it can be readily dismantled for internal cleaning. Refer to Part 4, Technical Specification M-001 for design conditions.

Bidder shall consider the cooling water temperature variations expected and shall provide a temperature control system (including control valve) to maintain proper oil temperature over the entire range of water temperatures.

4.8 **Fittings**

A non-return valve of approved design and manufacture shall be fitted at the discharge from each pump and adequate provision shall be made to avoid possible damage to pump and driver and equipment on the suction side of the pump due to possible failure of the non-return valve.

Manually operated suction isolating valves for each boiler feed pump shall be provided. The discharge valves shall be power operated.

4.9 **Automatic Recirculation**

A fail-safe (fail open) automatic minimum flow recirculation arrangement shall be provided to prevent damage to the pumps at low flows and during start-up. The discharge from each minimum flow device shall be individually run to the deaerator and provided with isolating valves. Means shall be provided for “on-load” testing of the leak-off devices.

A connection shall be provided on the pump discharge piping to recirculate water to the deaerator when operating between shutoff and very low capacities. The Bidder shall install a suitable pressure breakdown device and an on/off recirculation control valve complete with limit switches in the recirculation line.

4.10 **Strainers**

The feed pump suction pipework will have separate downcomers to each pump suction.

Each pressure circuit pumps shall be provided with one dual type strainer to protect the pump against following material.

4.11 **Pump Support and Bedplate**

A bedplate shall be used to support the pump and the motor as a complete unit. Assembly of the unit shall be at the manufacturer’s works.

The bedplate shall be manufacturer’s standard fabricated steel type.

Bidder shall supply all anchoring materials, including anchor bolts and templates. Sole plates, expansion keyways (or equivalent) and anchors shall be ruggedly fabricated, preferably in a single assembly (sole plate structure).

4.12 **Materials**

- | | | |
|---|--|---|
| - | Casing, suction | Carbon steel / Forged steel and discharge nozzles |
| - | Shaft | Stainless Steel |
| - | Impeller and internal stationery parts | Stainless Steel |

solutions used internally shall be easily removable. Inlets and outlets shall be blanked or plugged.

4.16 Noise

Noise of the assembled unit (motor and pump) shall be in accordance with Part 4, Subsection 2.9

5. PERFORMANCE TESTING

5.1 Shop Test

Shop test shall conform to the requirements of part 4, specification M 014.

Tests shall be conducted generally in accordance with the latest edition of ASME performance test code PTC 8.2, Centrifugal pumps.

Each pump and spare rotary assembly shall be shop tested by the Bidder follows:

A performance test consistent of developed head, efficiency, break power and NPSH required versus flow shall be performed on each pump at design operating speed and maximum water temperature.

A minimum of five test points shall be taken at full design speed and temperature. Flow shall range from pump minimum flow to not less than 25% beyond the design flow. One of the test points shall be at the design head and capacity.

NPSH test shall be carried out to determine the limiting suction requirements. Incipient cavitation shall be determined by a 1% decrease in pump head.

When the Bidder's facilities do not permit full speed tests, the bid proposal shall described in the detail which reduced speed shop test can be performed that representative of full speed design conditions.

Each pump shall be subjected to hydrostatic test 1.5 times the design pressure.

The shop test shall be carried out on the complete unit ; motor, pumps and speed increaser. Deviation from this document shall be stated in the bid proposal.

5.2 **Test Curves**

Although of design point is of primary importance in the guarantee, any pump is subject to rejection at the time the test curves are received unless they indicate the following :

The As- tested head capacity curve:

- Stable, steadily rising from the design point to shut off
- Rise-to-shutoff within specified limits. If no such limits are specified, it is understood that permissible variation of the percent rise-to-shutoff is plus or minus 15% of the bid proposal curve percent rise. For example if the curve rise-to shutoff is 20%, the permissible variation on the plus and minus side are 23% and 17%, respectively.

The As-Tested Power curve:

- Design point (guaranteed) power shall not be exceeded at any point from zero capacity to design capacity.
- Shall have values not exceeding the corresponding Bid proposal curve values in the runout area. As an alternative, so as to cover overshoot of head, the bid proposal curve may indicate a guaranteed maximum power at the design and guaranteed maximum hose power at runout.

The As- Tested NPSH curve

- Shall have values less than the design (guaranteed) NPSH at all capacities below design capacity
- Shall have values not exceeding the corresponding Bid proposal curve values in the runout area.

If the test curves are rejected based on deviation from the above requirement all corrective measure shall be at the Bidder's expense.

Reference to the recommendations of the “hydraulic Institute” does not excuse the Bidder from complying with the above requirements or other specified herein.

Any deficiency brought out by this test shall be promptly corrected by the Bidder at no cost to the Owner.

5.3 Site Test

After equipment is put in service, the Bidder shall conduct a running test to ascertain that the equipment is performing properly in every respect.

Any deficiency brought out by this test shall be promptly corrected by the Bidder at no cost to the Owner.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

6.1 Quality Control

Quality assurance/control shall conform to the requirements of Part 4, Subsection 2.7.

6.2 Examinations

In addition to Bidder’s normal quality control inspection procedures for ensuring high quality pumps, each pump shall be examined as follows. These are intended to be minimum requirements and not to limit or reduce Bidder’s standard practices or applicable code requirements if they are more stringent and broader in coverage.

Ultrasonic : Following heat treatment, 100% ultrasonic inspection shall be performed on the pump shaft.

Testing shall carried out in accordance with the requirement of ASTM or appropriate standard subject to Owner approval.

Radiography : Following heat treatment, 100% volumetric inspection shall be performed on any fabrication welds in the pump casing. Radiography shall be performed in accordance with the requirements of ANSI B31.1, or appropriate standard subject to Owner approval.

Magnetic particle or liquid penetrant : The following pump parts shall receive an examination by either Magnetic particle or liquid penetrant after final machining

- Impellers
- Volute casings
- Diffusers
- High pressure nozzle field weld and preparations
- Shaft

As a minimum requirement, all high potential stress areas shall be examined. These would include impeller inlet and exit ends, volute casings, diffuser vane inlet and exit ends, root attachment areas of diffuser vanes to the diaphragms. Areas examined shall extend a minimum of 50 mm from vane or cutwater edge measured along the flow path and 25 mm minimum on each side of diffuser vane root attachments.

Magnetic particle examination shall be performed on parts that are ferromagnetic. Procedure utilized shall meet the requirements of ASME V.

Liquid penetrant examination procedures shall meet the requirements of the ASTM E165 or ASME V.

In addition to the requirement set forth in paragraph above each shaft including bearing journals shall be examined by the Liquid Penetrant method.

- examination shall be conducted after the shaft and bearing journals have been finished, but before and plating or overlay is applied.

Acceptance Rejection Criteria:

- Ultrasonic (pump shaft) : Refer to ASTM or appropriate standard subject to Owner approval. The Bidder shall propose the highest criteria level for this inspection.
- Radiography : Discontinuities exceeding the acceptance standards of Chapter VI of ANSI B31.1 shall be unacceptable.

- Magnetic Particle

The Bidder shall refer to ASTM or appropriate standard. The Bidder shall propose the highest criteria for this inspection.

Forgings and Wrought Materials : Indications exceeding acceptance standards of Chapter VI of ANSI B31.1 shall be - unacceptable.

- Liquid Penetrant : Same as magnetic particle.

7. INSTRUMENTATION AND CONTROLS

7.1 General

The Bidder shall be responsible for the design, supply and installation of instrumentation and control devices as specified herein. The Bidder shall also be responsible for the complete design, supply and installation of all local instrumentation and control systems, devices, test instruments and panels integral with the HRSG feed pumps whether or not they are specified herein.

The HRSG feed pump shall be controlled and monitored on the Operator Station in the Central Control Room.

Refer to Part 4, Technical Specification I-001 Instrumentation and Controls for design requirements for the instrumentation and control devices.

Refer to Part 4, Technical Specification E-001, General Electrical Specification for design requirements for the junction boxes, cables and wiring.

7.2 Local Control Panels

The Bidder shall furnish and install a local Control Panel for the Pumps. The local control panel will only be used for emergency maintenance and testing purposes. Refer to Part 4, Technical Specification I-002 for design requirement for Local Control Panels.

7.3 Facilities for Determining Speed and Vibration

Both the pump and the driver shall have the shaft end opposite the coupling available for taking readings with a portable tachometer.

Vibration Measuring Devices

- Bidder shall provide vibration measuring devices mounted for each pressure part of HRSG Feed Pump, as below. The vibration monitor shall be in accordance with Part 4, Technical Specification I-001. Each pump shall be provided with four radially mounted sensors and one axially mounted sensor. Each motor shall be provided with two radially mounted sensors.

7.4 **Thermocouples**

All bearings of each pump shall be furnished with a thermowell and thermocouple as reviewed by the Owner. Thermocouple wells shall be embedded in the metal as close as possible to the sleeve bearing surfaces or to the antifriction races and shall be located in the bottom half of bearing, in each case. The thrust bearing thermocouple shall be in the oil stream leaving the thrust bearing.

7.5 **Pump Protection**

Equipment shall be provided to trip an operational HRSG feed pump in the event of:

- Extreme low level in the deaerator
- Low suction pressure arising from suction valve closure or a blocked suction

Each of the trip features shall be alarmed and annunciators shall be provided in the central control room in accordance with the relevant section of this specification.

8. **ELECTRICAL SPECIFICATIONS**

8.1 **General**

The Bidder shall furnish and install all electrical equipment and accessories required for the HRSG Feed Pumps.

For general electrical requirements, see Part 4, Technical Specification E-001.

8.2 **Motors**

For detailed electrical requirements, see Part 4, Technical Specification E-210 for Low Voltage Motors, E-211 for Medium Voltage Motors.

The Bidder shall submit the method of starting equipment for the motor to the Owner for approval.

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